SIMATS ENGINEERING

ASSESSMENT TOOL 3: Correlation and Pattern Analysis

COURSE NAME: Query Processing for Data Science **COURSE CODE:** DSA0503

COURSE OUTCOME COVERED

CO4: Explore datasets using analytical techniques such as grouping, correlation detection, joining datasets, and extracting meaningful insights.

Assessment Tool Weightage: 25%

Total Marks: 20

Code of Conduct

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: ____________________

Correlation and Pattern Analysis Rubric

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Data Importing, Exploration & Preparation	25	Thorough understanding; correctly imports, explores, cleans, and prepares data;	Good understanding; performs most data preparation tasks correctly with minor issues	Basic understanding; performs some preparation tasks but misses important issues	Poor understanding; unable to import, explore, or prepare the data correctly
Python Coding & Data Analysis	25	Effectively applies relevant Python concepts; code is accurate, efficient, and performs analysis appropriately	Applies appropriate Python concepts with minor coding or analytical gaps	Limited application of Python concepts; analysis is partially completed	Fails to apply relevant Python concepts; major errors or incomplete analysis
Correlation Analysis & Visualization	20	Performs in-depth correlation analysis with appropriate visualizations; demonstrates strong reasoning and insights	Performs good correlation analysis with suitable visualizations and logical reasoning	Performs basic correlation analysis with limited visualization or interpretation	Weak or no correlation analysis; inappropriate/missing visualizations and lacks reasoning
Pattern Identification & Interpretation	20	Identifies meaningful and insightful patterns; provides innovative, feasible interpretations with strong justification	Identifies relevant patterns and provides logical interpretations with adequate justification	Identifies basic patterns with weak or limited justification	Incorrect or no meaningful patterns identified; interpretation is missing or inappropriate
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Total Marks Awarded:

Signature of the Course Faculty

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1. Student Performance Pattern Analysis

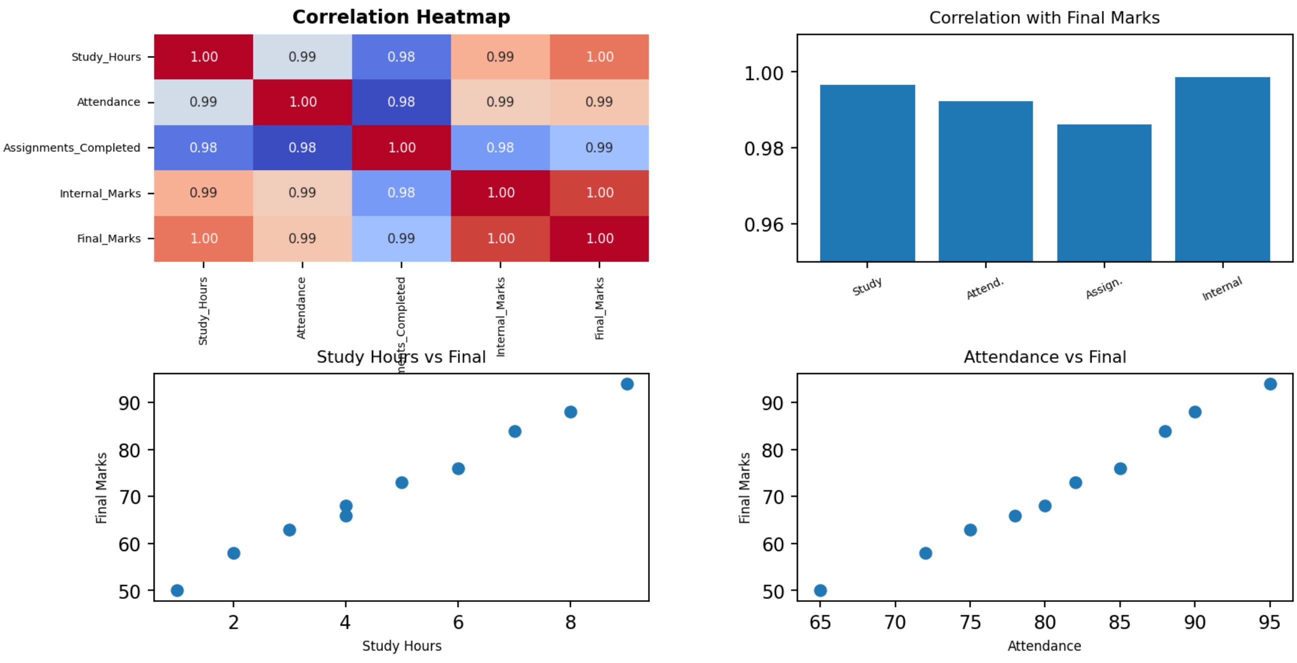
A college has collected the following data for students: Student_ID, Study_Hours, Attendance, Assignments_Completed, Internal_Marks, Final_Marks. Write a Python program using Pandas to import the dataset, explore its structure and summary statistics, calculate correlations between Study Hours/Attendance/Assignments Completed/Internal Marks and Final Marks, display a correlation matrix using a heatmap, identify the strongest relationship with Final Marks, and describe at least two student-performance patterns.

Code

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

data = {
"Student_ID":["S01","S02","S03","S04","S05","S06","S07","S08","S09","S10"],
"Study_Hours":[2,4,6,8,3,9,5,1,7,4],
"Attendance":[72,80,85,90,75,95,82,65,88,78],
"Assignments_Completed":[5,7,8,10,6,10,8,4,9,6],
"Internal_Marks":[55,65,72,82,60,90,70,48,80,62],
"Final_Marks":[58,68,76,88,63,94,73,50,84,66]}
df = pd.DataFrame(data)

print(df.head())
print(df.info())
print(df.describe())

corr = df.select_dtypes("number").corr()
print("\nRequired correlations:")
for col in ["Study_Hours","Attendance","Assignments_Completed","Internal_Marks"]:
    print(col, "vs Final Marks:", round(corr.loc[col,"Final_Marks"],3))

sns.heatmap(corr, annot=True, cmap="coolwarm", fmt=".2f")
plt.title("Student Performance Correlation Matrix")
plt.show()

target = corr["Final_Marks"].drop("Final_Marks")
print("Strongest relationship:", target.abs().idxmax())
```

Output

Google Colab Output — Student Performance




```
Study Hours ↔ Final Marks       r = 0.997
Attendance ↔ Final Marks        r = 0.992
Assignments ↔ Final Marks       r = 0.986
Internal Marks ↔ Final Marks     r = 0.999
Strongest relationship: Internal Marks (r = 0.999)
```

2. Sales and Customer Behavior Pattern Analysis

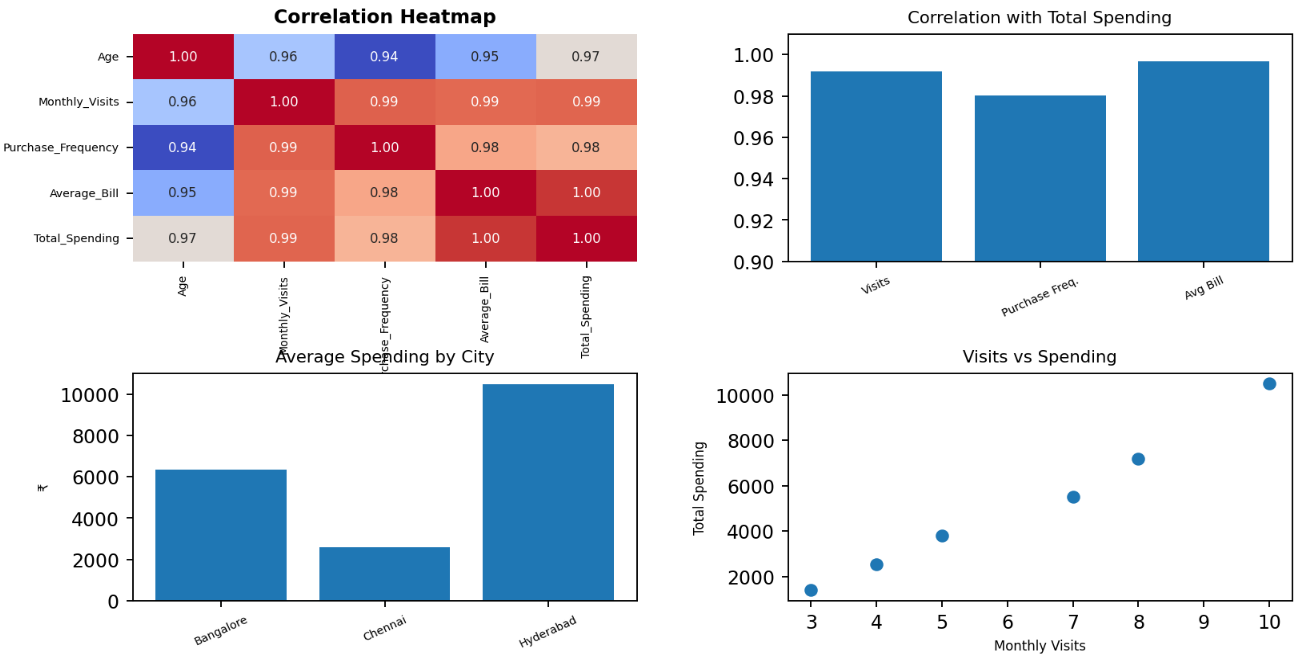
A retail store maintains customer_data.csv and sales_data.csv. Write a Python program to import both CSV files, explore columns/data types/summary statistics, join them using Customer_ID, calculate correlations between Monthly Visits/Purchase Frequency/Average Bill and Total Spending, create a correlation heatmap, group customers by city and calculate average Total Spending, and identify purchasing-behavior patterns.

Code

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from io import StringIO

customer_csv = """Customer_ID,Age,City
C01,22,Chennai
C02,35,Bangalore
C03,28,Chennai
C04,42,Hyderabad
C05,31,Bangalore
C06,25,Chennai"""

sales_csv = """Customer_ID,Monthly_Visits,Purchase_Frequency,Average_Bill,Total_Spending
C01,4,3,850,2550
C02,8,6,1200,7200
C03,5,4,950,3800
C04,10,7,1500,10500
C05,7,5,1100,5500
C06,3,2,700,1400"""

customers = pd.read_csv(StringIO(customer_csv))
sales = pd.read_csv(StringIO(sales_csv))

print(customers.info())
print(sales.describe())

df = customers.merge(sales, on="Customer_ID")
corr = df.select_dtypes("number").corr()
print("\nRequired correlations:")
for col in ["Monthly_Visits","Purchase_Frequency","Average_Bill"]:
    print(col, "vs Total Spending:", round(corr.loc[col,"Total_Spending"],3))

sns.heatmap(corr, annot=True, cmap="coolwarm", fmt=".2f")
plt.title("Customer Behavior Correlation Matrix")
plt.show()

city_avg = df.groupby("City")["Total_Spending"].mean()
print("\nAverage spending by city:")
print(city_avg)
```

Output

Google Colab Output — Sales & Customer Behavior




```
Monthly Visits ↔ Total Spending     r = 0.992
Purchase Frequency ↔ Spending       r = 0.980
Average Bill ↔ Total Spending       r = 0.997
Average spending: Bangalore ₹6350 | Chennai ₹2583 | Hyderabad ₹10500
Strongest relationship: Average Bill with Total Spending
```

3. Weather Data: Identifying Correlation and Temperature Patterns

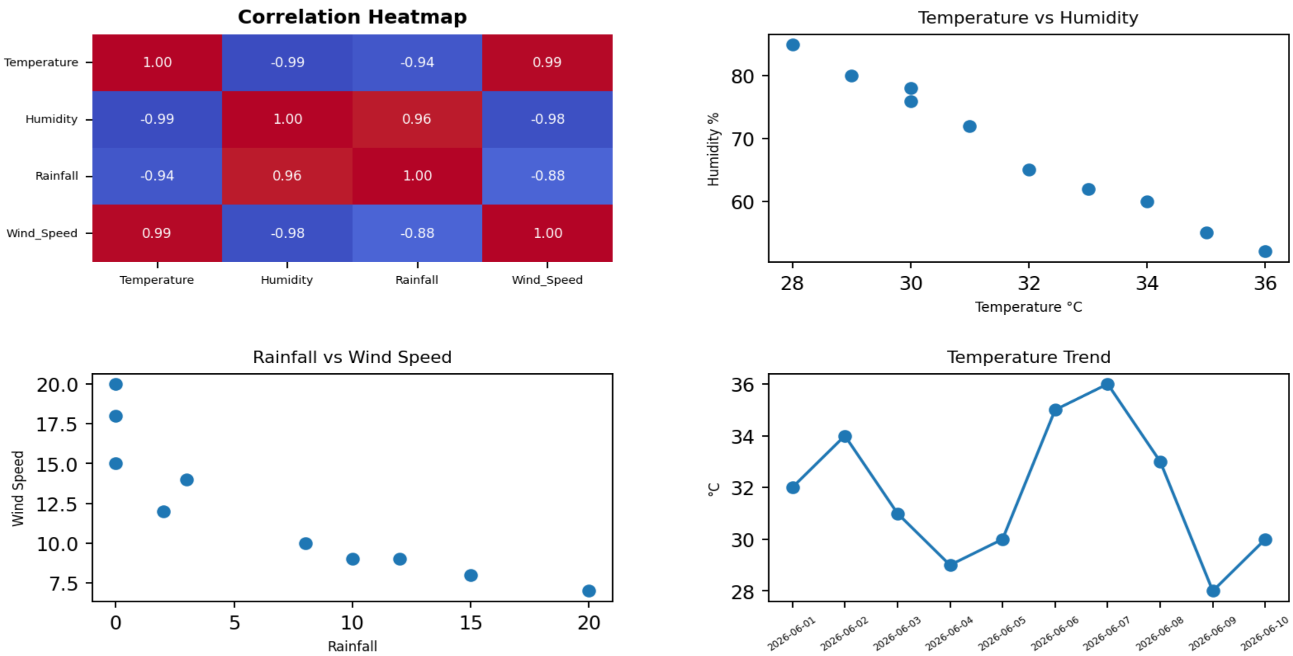
A weather station records Date, Temperature, Humidity, Rainfall and Wind Speed. Using Python, Pandas, Matplotlib and Seaborn, import and inspect the dataset, convert Date to an appropriate format, calculate and visualize the numerical correlation matrix, determine the Temperature-Humidity relationship, identify the Rainfall-Wind Speed relationship, plot Temperature against Humidity, and identify at least three weather patterns.

Code

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from io import StringIO

csv_data = """Date,Temperature,Humidity,Rainfall,Wind_Speed
01-06-2026,32,65,2,12
02-06-2026,34,60,0,15
03-06-2026,31,72,8,10
04-06-2026,29,80,15,8
05-06-2026,30,78,12,9
06-06-2026,35,55,0,18
07-06-2026,36,52,0,20
08-06-2026,33,62,3,14
09-06-2026,28,85,20,7
10-06-2026,30,76,10,9"""

df = pd.read_csv(StringIO(csv_data))
df["Date"] = pd.to_datetime(df["Date"], dayfirst=True)

print(df.head())
print(df.info())
corr = df.select_dtypes("number").corr()
print("\nCorrelation Matrix:")
print(corr.round(3))

sns.heatmap(corr, annot=True, cmap="coolwarm", fmt=".2f")
plt.title("Weather Correlation Matrix")
plt.show()

print("Temperature-Humidity:", round(corr.loc["Temperature","Humidity"],3))
print("Rainfall-Wind Speed:", round(corr.loc["Rainfall","Wind_Speed"],3))

plt.scatter(df["Temperature"], df["Humidity"])
plt.xlabel("Temperature")
plt.ylabel("Humidity")
plt.title("Temperature vs Humidity")
plt.show()
```

Output

Google Colab Output — Weather Correlation




```
Temperature ↔ Humidity       r = -0.993 (negative)
Rainfall ↔ Wind Speed         r = -0.883 (negative)
Temperature ↔ Rainfall        r = -0.940 (negative)
Temperature ↔ Wind Speed      r = 0.988 (positive)
Pattern: hotter days generally have lower humidity.
```

4. Employee Productivity Pattern Analysis

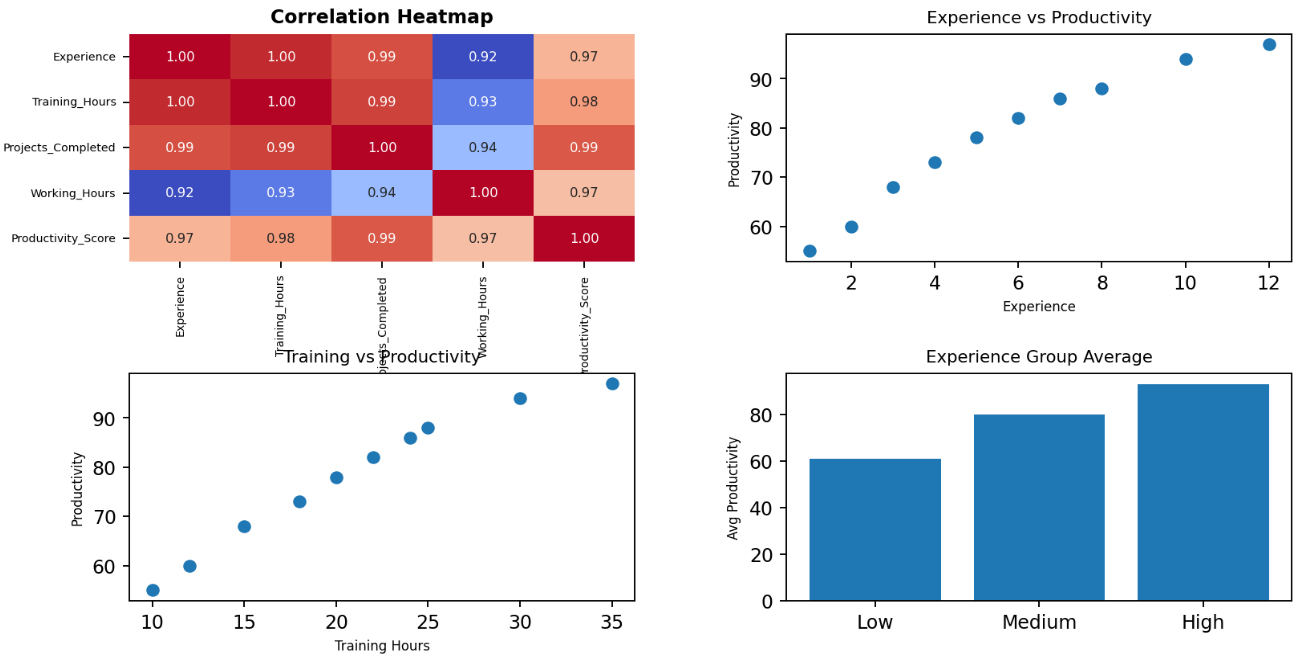
An organization wants to understand factors affecting employee productivity using Employee_ID, Experience, Training_Hours, Projects_Completed, Working_Hours and Productivity_Score. Develop a Python program to perform exploratory analysis, calculate correlations with Productivity Score, visualize a heatmap, create scatter plots for the two strongest variables, group employees into low/medium/high experience groups, calculate average productivity, and conclude whether experience and training are associated with productivity.

Code

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

data = {
"Employee_ID":["E01","E02","E03","E04","E05","E06","E07","E08","E09","E10"],
"Experience":[1,3,5,8,2,10,6,4,12,7],
"Training_Hours":[10,15,20,25,12,30,22,18,35,24],
"Projects_Completed":[2,4,6,8,3,10,7,5,11,8],
"Working_Hours":[35,40,42,45,38,44,43,41,46,44],
"Productivity_Score":[55,68,78,88,60,94,82,73,97,86]}
df = pd.DataFrame(data)

print(df.info())
print(df.describe())

corr = df.select_dtypes("number").corr()
print("\nProductivity correlations:")
for col in ["Experience","Training_Hours","Projects_Completed","Working_Hours"]:
    print(col, "vs Productivity:", round(corr.loc[col,"Productivity_Score"],3))

sns.heatmap(corr, annot=True, cmap="coolwarm", fmt=".2f")
plt.title("Employee Productivity Correlation Matrix")
plt.show()

for col in ["Projects_Completed","Training_Hours"]:
    plt.scatter(df[col], df["Productivity_Score"])
    plt.xlabel(col)
    plt.ylabel("Productivity Score")
    plt.title(f"{col} vs Productivity")
    plt.show()

df["Experience_Group"] = pd.cut(
    df["Experience"], bins=[0,3,7,99],
    labels=["Low","Medium","High"])

print("\nAverage Productivity by Experience Group:")
print(df.groupby("Experience_Group", observed=True)["Productivity_Score"].mean())
```

Output

Google Colab Output — Employee Productivity




```
Experience ↔ Productivity       r = 0.973
Training Hours ↔ Productivity    r = 0.979
Projects ↔ Productivity          r = 0.991
Working Hours ↔ Productivity     r = 0.971
Strongest: Projects Completed (r = 0.991)
Average Productivity: Low 61.00 | Medium 79.75 | High 93.00
```

5. E-Commerce Dataset: Discovering Hidden Purchase Patterns

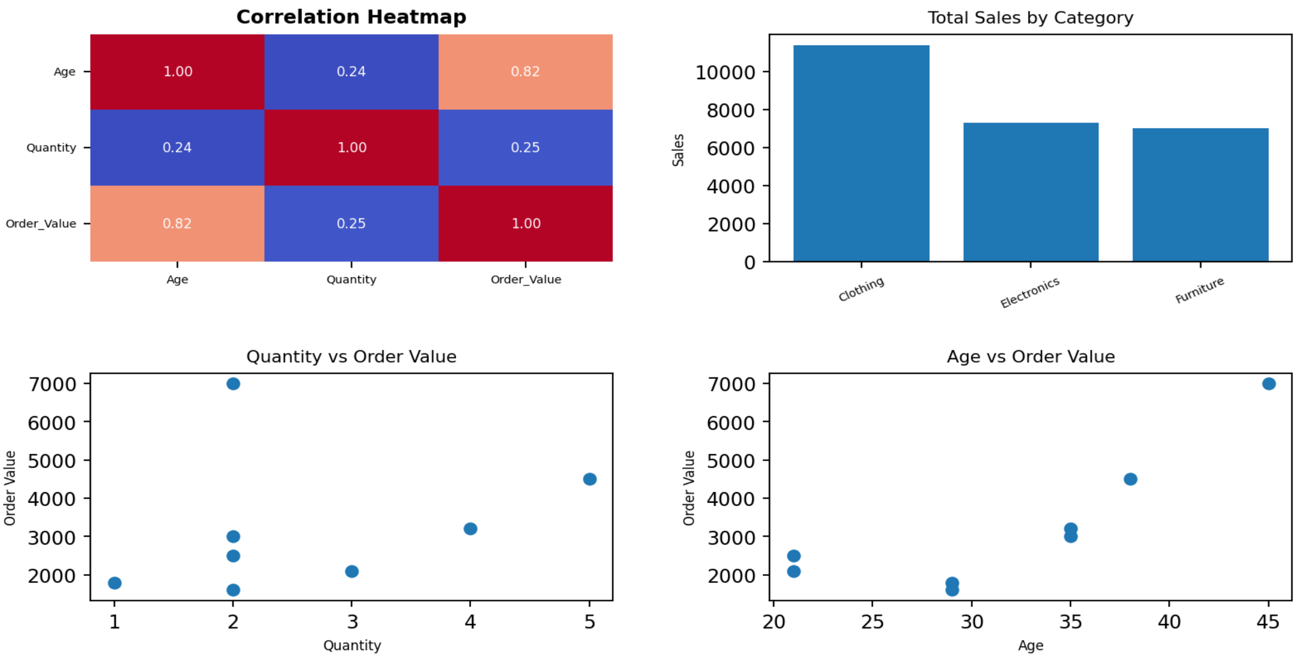
An e-commerce company has customers.csv and orders.csv. Write a Python program to import and explore the datasets, join them using Customer_ID, calculate correlations between Age/Quantity and Order Value, group orders by Product Category to calculate total quantity, total order value and average order value, identify the highest-sales category, create suitable visualizations, determine whether quantity is strongly related to order value, and identify at least three meaningful purchase patterns.

Code

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from io import StringIO

customers_csv = """Customer_ID,Age,Gender,City
C01,21,M,Chennai
C02,35,F,Bangalore
C03,29,M,Chennai
C04,45,F,Hyderabad
C05,38,M,Bangalore"""

orders_csv = """Order_ID,Customer_ID,Product_Category,Quantity,Order_Value
O01,C01,Electronics,2,2500
O02,C02,Clothing,4,3200
O03,C03,Electronics,1,1800
O04,C04,Furniture,2,7000
O05,C05,Clothing,5,4500
O06,C01,Clothing,3,2100
O07,C02,Electronics,2,3000
O08,C03,Clothing,2,1600"""

customers = pd.read_csv(StringIO(customers_csv))
orders = pd.read_csv(StringIO(orders_csv))

print(customers.info())
print(orders.describe())

df = customers.merge(orders, on="Customer_ID")
corr = df[["Age","Quantity","Order_Value"]].corr()
print("\nAge vs Order Value:", round(corr.loc["Age","Order_Value"],3))
print("Quantity vs Order Value:", round(corr.loc["Quantity","Order_Value"],3))

summary = df.groupby("Product_Category").agg(
    Total_Quantity=("Quantity","sum"),
    Total_Order_Value=("Order_Value","sum"),
    Average_Order_Value=("Order_Value","mean"))

print("\nCategory Summary:")
print(summary)

sns.heatmap(corr, annot=True, cmap="coolwarm", fmt=".2f")
plt.title("E-Commerce Correlation Matrix")
plt.show()

summary["Total_Order_Value"].plot(kind="bar")
plt.title("Total Sales by Product Category")
plt.ylabel("Total Order Value")
plt.show()

plt.scatter(df["Quantity"], df["Order_Value"])
plt.xlabel("Quantity")
plt.ylabel("Order Value")
plt.title("Quantity vs Order Value")
plt.show()

print("Highest total sales:", summary["Total_Order_Value"].idxmax())
```

Output

Google Colab Output — E-Commerce Purchase Patterns




```
Age ↔ Order Value            r = 0.823
Quantity ↔ Order Value        r = 0.254
Highest total sales: Clothing = ₹11,400
Category totals: Clothing 14 units / ₹11,400 | Electronics 5 / ₹7,300 | Furniture 2 / ₹7,000
Quantity is only weakly related to order value in this sample.
Pattern: Clothing has the highest total sales and quantity sold.
```